

VariCAD: 3D/2D CAD Software

Apart from increasing the productivity of designers and improving the quality of design, computer-aided design (CAD) software assist in the creation, analysis, modification or optimisation of a design. The major areas where CAD is used are electronic design, mechanical design and engineering drawings. This month's EFY Plus DVD includes VariCAD. Here is an overview of this software

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VariCAD is a software for 3D/2D CAD that runs on Windows and Linux. In addition to tools for 3D modelling and 2D drawing-cum-editing, it provides support for geometrical constraints, tools for shells, pipelines, sheet metal unbending and crash tests, and assembly support to name a few. The system contains both 3D and 2D libraries of mechanical parts (DIN and ANSI), hydraulic and electrical symbols, and tolerance symbols as well.

Impressive functionality

VariCAD features a lot of interesting functions that vary from an intuitive user interface (UI) all the way to performing virtual crash tests and creating a bill of materials (BOM).

User interface and system environment. Designed to allow quick and intuitive 3D/2D orientation, VariCAD's graphical user interface has been carefully tailored and tuned to reflect the thought process of a designer so that ideas can be captured and communicated with a minimal number of steps. The commands have been created with a focus on ease of use.

It is possible to start by creating a 3D model and then use it to automatically create drawing files, or draw only in 2D. Designing in 3D is generally more natural since it closely represents actual parts and assemblies. Hence the 3D approach is usually more intuitive than 2D



Fig. 1: VariCAD user interface

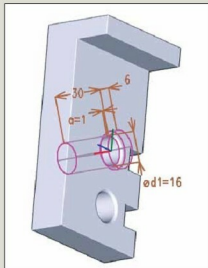


Fig. 2: Definition of parameters and geometric constraints

drafting. Models created in 3D can be easily converted into conventional 2D documentation.

3D modelling. With the help of a library of basic 3D shapes (box, cylinder,

cone, etc) that could be easily modified by editing their dimensions, solids can also be created by profile rotation, lofting or extrusion. Tools with higher complexity include rotation blending between two profiles, creation of helical surfaces, and lofting between a circle and rectangle, or between different profiles.

Addition or subtraction of solids is possible, thus forming Boolean trees representing real mechanical parts. Automatic trimming is available for Boolean operations. Predefined operations like groove milling, face milling or drilling of holes are also available. With an option of rounded or chamfered edges, VariCAD provides a lot of possibilities of solid transformations or their editing. Also, Boolean trees can be easily edited by selecting them from a list displaying the structure, or by selecting solid parts from 3D.

Parameters and geometrical constraints. Solids or their parts can be comfortably transformed, and it is also possible to optionally define geometrical constraints. Once defined, these constraints allow the user to 'stick' object at the defined location. Constrained object changes its position automatically if other objects are changed or transformed. For example, if a user constrains a groove to the end